Computer-Based Data Processing

Lab 2 – Pivot Tables with LibreOffice Calc



Objectives

In the previous lab, we used formulas to work directly with the rows of a dataset.

We will now focus on another question:

How can we quickly summarise hundreds or thousands of rows?

To do this, we will use **pivot tables** in LibreOffice Calc.

Throughout this lab, we will work with a dataset containing tweets posted around New Year 2015, describing their authors' New Year's resolutions.

By the end of this lab, you should be able to:

- create a pivot table;
- count observations by category;
- cross-tabulate several variables;
- display results as percentages;
- filter and sort results;
- examine a distribution over time;
- create a chart from a pivot table.

Start by downloading the `new_years_resolutions.csv` file from the course platform, then open it with LibreOffice Calc.

Important: as in the previous lab, check that LibreOffice Calc uses English locale settings. Go to `Tools > Options > Languages and Locales > General`, then select `English (USA)` under `Locale setting`.

1 Exploring the dataset

Each row in the table corresponds to a tweet.

The available information includes:

- the date and time of the tweet;
- the original text;
- the resolution category;
- a more specific topic associated with the resolution;
- the user's location;
- their US state;
- their region;
- their time zone;
- their gender.

Take a few minutes to explore the table.

Pay particular attention to the columns:

`tweet_category`, `tweet_topics`, `tweet_state`, `tweet_region`

as well as the column containing the date and time of the tweet.

We will explore what these data tell us about the resolutions people made for 2015.

2 Your first pivot table

First, we want to find out which resolution categories are the most common.

Select the table, then create a new pivot table from the data.

In this pivot table:

- place `tweet_category` in the row fields;
- also use `tweet_category` as a data field;
- choose a calculation that **counts** the number of tweets.

You should obtain a table showing the number of tweets for each resolution category.

Sort the results from the most frequent category to the least frequent.

What to look for: which categories seem particularly prominent among New Year's resolutions?

3 From counts to proportions

Raw counts allow us to compare categories, but a distribution is often easier to interpret when expressed as percentages.

Modify your pivot table to display each category's share of all tweets.

The total should be:

100 %.

Compare these results with the counts from the previous section.

Note: the same information can be presented in several ways without changing the original data.

4 Cross-tabulating several variables

A pivot table becomes particularly useful when you want to examine several variables simultaneously.

Modify your table to compare resolution categories across geographical regions.

Use:

- `tweet_category` as the row field;
- `tweet_region` as the column field;
- the number of tweets as the value.

You now have a two-dimensional table.

Compare the different regions:

- do some categories seem more common in one region than in another?
- are the observed differences easy to interpret using counts alone?

Then try swapping the row and column fields.

The table contains exactly the same information, but the results are read differently.

5 Adding a filter

We now want to be able to explore only part of the dataset.

Add `user_gender` as a filter field in the pivot table.

Display the following in turn:

- all tweets;
- only tweets associated with users labelled `female`;
- only tweets associated with users labelled `male`.

Observe how the table updates when the filter changes.

You can also try using `tweet_region` as a filter.

6 At what time did people tweet their resolutions?

The `tweet_created` field contains both the date and the time of publication.

We want to examine only the **hour of the day**.

In the original table, add a new column called:

`Hour`

In the first data row, extract the hour from the tweet's creation date and time.

Depending on your LibreOffice Calc configuration, the function used to extract the hour may appear as:

```
=HOUR(A2)
```

or under its localised name.

Then fill the formula down through the entire dataset.

> **Tip:** as in the previous lab, you can double-click the fill handle to quickly apply the formula to all rows.

Create a new pivot table in which:

- the hour appears in the row fields;
- the number of tweets appears as the value.

This gives you the distribution of tweets throughout the day.

Sort the hours in chronological order, from 0 to 23.

7 Visualising the results

A table is useful for reading exact values, but a chart often makes a trend easier to identify.

Using the previous table, create a chart showing the number of tweets by hour.

Choose a chart type suitable for a discrete variable such as the hour of the day.

Your chart should make it easy to identify:

- periods when few tweets are posted;
- the most active periods;
- the hour with the highest number of tweets.

Give your chart a descriptive title.

8 Where do the tweets come from?

We will now explore the geographical dimension.

Create a new pivot table showing the number of tweets for each value of:

`tweet_state`

Sort the states from the highest to the lowest number of tweets.

> **What to look for:** do some states stand out clearly from the others?

Then create a chart comparing the states with the most tweets.

You do not need to display every state if this makes the chart difficult to read: you can limit it to the states with the highest counts.

9 Exploring topics in more detail

The `tweet_category` column provides a broad category.

The `tweet_topics` column contains a more detailed description of the resolution.

Create a pivot table to examine the topics within a particular category.

For example, you can:

- use `tweet_category` as a filter;
- place `tweet_topics` in the row fields;
- count the number of tweets.

Choose a category you find interesting, then examine the subtopics it contains.

This step illustrates how a pivot table allows you to move quickly from a **general overview** to a **more detailed view**.

Further practice

If you have finished the lab, use the variables in the dataset to answer a question of your choice.

Here are a few examples:

- are resolution categories the same across regions?
- are some categories more common among men or women?
- do the topics discussed vary according to the time of publication?
- which region has the highest proportion of tweets about health and fitness?

Build the pivot table that you think is best suited to your question.

If appropriate, also present the results in a chart.

Review

During this lab, you used pivot tables to summarise and explore a dataset.
You learned how to:

- automatically count observations by category;
- display results as proportions;
- cross-tabulate several variables;
- filter a pivot table;
- reorganise a pivot table without changing the data;
- examine a distribution by hour;
- sort results;
- create a chart from summarised data.